NAME

ovn-br-controller – Open Virtual Network local OVS bridge controller

SYNOPSIS

ovn−br−controller [*options*] [*ovs-database*]

DESCRIPTION

ovn−br−controller is OVN logical flow based local controller daemon to control and program the Open vSwitch bridges. It connects up to the OVN bridge database (see **ovn−br**(5)) over the OVSDB protocol, and down to the Open vSwitch database (see **ovs−vswitchd.conf.db**(5)) over the OVSDB protocol and to **ovs−vswitchd**(8) via OpenFlow. **ovn−br−controller** processes and converts the OVN Logical flows configured in the OVN bridge database for the Open vSwitch bridges to OpenFlow rules and programs these flow rules to the Open vSwitch bridges.

CONFIGURATION

ovn−br−controller retrieves most of its configuration information from the local Open vSwitch's ovsdb-server instance. The default location is **db.sock** in the local Open vSwitch's "run" directory. It may be overridden by specifying the *ovs-database* argument as an OVSDB active or passive connection method, as described in **ovsdb**(7).

ovn−br−controller assumes it gets configuration information from the following keys in the **Open_vSwitch** table of the local OVS instance:

external_ids:ovn−br−remote
: The OVN bridge database that this system should connect to for its configuration, in one of the same forms documented above for the *ovs-database*.

ovn−br−controller reads the following values from the **Open_vSwitch** database of the local OVS instance:

external_ids:ovnbr−managed from **Interface** table
: If the value is **true**, then this interface is managed by **ovn−br−controller** if this interface is connected to the OVS bridge managed by **ovn−br−controller**.

  Default value is true if this option is not defined.

  CMS should set the value of this option to **false** if it doesn't want **ovn−br−controller** to manage it.

RUNTIME MANAGEMENT COMMANDS

ovn−appctl can send commands to a running **ovn−br−controller** process. The currently supported commands are described below.

inc−engine/show−stats
: Display **ovn−br−controller** engine counters. For each engine node the following counters have been added:

  - **recompute**
  - **compute**
  - **cancel**

inc−engine/show−stats *engine_node_name* [*counter_name*]
: Display the **ovn−br−controller** engine counter(s) for the specified *engine_node_name*. *counter_name* is optional and can be one of **recompute**, **compute** or **cancel**.

inc−engine/clear−stats
: Reset **ovn−br−controller** engine counters.

inc−engine/recompute
: Triggers a full recompute of the incremental processing engine on the next iteration.

inc−engine/compute−log−timeout *msecs*
Sets the timeout in milliseconds for logging engine compute events.

inc−engine/list−stopwatches [*node*]
Lists the engine nodes and their change handler names. If *node* is specified, only the handlers for that engine node are listed.

inc−engine/enable−stopwatch [*node*]
Enables the per-node and per-handler stopwatches in the incremental processing engine. If *node* is specified, only that node's stopwatches are enabled; otherwise all nodes are affected. While enabled, every engine node recompute and change handler invocation is timed and the results can be viewed with the **stopwatch/show** command. Stopwatches are disabled by default because they add measurable CPU overhead.

inc−engine/disable−stopwatch [*node*]
Disables the per-node and per-handler stopwatches in the incremental processing engine. If *node* is specified, only that node's stopwatches are disabled; otherwise all nodes are affected. This is the default state.